

Parent-Level Debt and Subsidiary-Level Risk-Taking: Evidence from Large Bank Holding Companies *

Kebin Ma[†] David R. Skeie[‡] Jianlin Wang[§] Jing Ye[¶]

April 19, 2026

Abstract

We investigate the influence of the parent company’s capital structure of a bank holding company’s (BHC) on the risk-taking behavior of its subsidiary banks, taking into account the BHC’s organizational structure. Our analysis shows that increased parent debt is associated with reduced risk-taking by subsidiary banks. This effect is particularly prominent in BHCs with more than one subsidiary bank or significant nonbank business, which would give the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank’s risk profile.

Keywords: bank holding company, parent-level debt, bank subsidiary, risk management

JEL Codes: G21, G28, G32, G33, G38

*We thank Yun Wang (CTFM discussant), Rebel A. Cole (AFBC discussant), and seminar participants at the 38th Australasian Finance and Banking Conference (2025), Contemporary Topics on Financial Markets (2025), and the 2024 Warwick Business School Brown Bag Series for helpful comments and feedback at various stages of this project. We also thank the staff at the Federal Reserve Banks across different districts for providing the FR Y-6 Annual Report of Holding Companies.

[†]Warwick Business School, University of Warwick. E-mail: kebin.ma@wbs.ac.uk

[‡]Warwick Business School, University of Warwick. E-mail: david.skeie@wbs.ac.uk

[§]Leavey School of Business, Santa Clara University. E-mail: jianlin.wang@scu.edu

[¶]Warwick Business School, University of Warwick. E-mail: jing.ye@warwick.ac.uk

1 Introduction

Since [Jensen and Meckling \[1976\]](#), corporate finance has a long tradition of considering the incentive implications of capital structure; the study of commercial banks makes no exception. Numerous empirical studies examine the impact of banks’ capital structure on their risk-taking incentives, and virtually all studies explaining bank risk-taking would include bank capitalization as a control variable. Yet most modern commercial banks operate within a bank holding company (BHC), raising an essential question: which entity’s capital position matters? A BHC can have several relevant balance sheets: the commercial bank’s own, those of other subsidiaries, the parent company’s, and the consolidated balance sheet of the whole group’s. While prior research typically focuses on the capitalization of the commercial bank *or* that of the consolidated BHC, this paper argues that one needs to take a wholistic view, emphasizing the capital position of the parent company. Specifically, we study how the *parent debt ratio*, defined as the share of a BHC’s non-deposit debt issued by its parent company, affects the risk-taking of its commercial bank subsidiaries.

[Avraham et al. \[2012\]](#) provide an insightful overview of the organizational structure of US bank holding companies. Figure 1 reproduces their findings for a typical large BHC, in which the bank holding company simultaneously controls a bank subsidiary, a non-bank subsidiary, and a commercial bank.¹ This multi-tiered organizational structure corresponds to a multi-tiered capital structure. Furthermore, the multi-level capital structure, as proxied by the allocation of external debt, also shows significant time-series and cross-sectional variation: between 2012 and 2022, the average parent debt ratio rose from 4.9% to 12.6% (Figure 2). In 2021, 33 of the 123 BHCs in our sample had no

¹Some BHCs also own foreign bank and non-bank subsidiaries.

parent-issued debt, while in the top half of the distribution the average parent debt ratio was 29.65%. Despite such variation, the effects of parent-level debt on subsidiary bank behavior remain unexplored. We address this gap by examining how the parent debt ratio influences subsidiary banks' risk-taking—an important question given that commercial banks account for the bulk of BHC assets, rising from 89.03% in 2012Q1 to 94.45% in 2022Q4 (Figure 3).

We find that a higher parent debt ratio is associated with decreased risk-taking by subsidiary banks. Specifically, an increase in the parent debt ratio correlates with lower risk-weighted leverage (measured as risk-weighted assets to tier 1 capital), reduced leverage, and a higher subsidiary bank z-score. These findings suggest that higher levels of parent debt ratio can lead to reduced risk-taking within subsidiary banks.

The underlying mechanism can be described as follows. When the parent company raises external debt and channels the proceeds into its subsidiary banks as equity, the resulting increase in capital constrains the subsidiary banks' incentives to take on risk, since potential losses are more fully absorbed by this equity buffer. Therefore, this parent-level debt effectively serves as a contractual commitment that helps mitigate risk-taking at the subsidiary level. In contrast, if the parent company issues less debt, subsidiary banks may need to rely more heavily on their own debt issuance to finance investments, which can encourage greater risk-taking due to limited liability. Under limited liability, downside losses are partly borne by debt holders, while shareholders retain the upside gains.

Similar reasoning extends to the case where the parent issues little debt but there is cross-guarantee. The cross-guarantee provision of Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA) permits the Federal Deposit Insurance Corporation (FDIC) to charge off any expected losses related to the failure of one sub-

subsidiary bank of a multi-bank holding company to the capital of a related subsidiary bank within the same multi-bank holding company ([Ashcraft \[2005\]](#)). This implies that BHCs with multiple subsidiary banks may engage in less risk-taking, as they must use their own capital to rescue failing subsidiary banks before FDIC intervention. Our findings confirm that subsidiary banks affiliated with multi-bank BHCs take on less risk when the parent company issues more debt.

Our paper highlights the function of internal capital markets within bank holding companies (BHCs). Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., [Cetorelli and Goldberg \[2012\]](#))—we emphasize the role of internal capital markets in shaping the risk-taking incentives of domestic subsidiary banks within U.S. BHCs.

While the empirical evidence is suggestive, a key concern in identifying the effect of parent debt on a subsidiary bank’s risk-taking is omitted variable bias. Specifically, when a subsidiary bank faces stringent regulatory capital requirements, it may reduce its risk-taking to avoid potential capital losses. At the same time, the parent company may issue additional external debt and inject the proceeds into the subsidiary bank to help it meet these requirements. In this case, the observed negative relationship is likely driven by the omitted factor—namely, the tightening of regulatory capital constraints—rather than a direct causal effect of parent debt.

To address this identification concern, we use, as a natural experiment, the COVID-19 outbreak in 2020Q1, which led to an exogenous increase in parent debt issuance. Under this identification, after 2020Q1, the reason for parent debt issuance is not endogenous but instead exogenous. Following the COVID-19 outbreak, parent debt issuance was driven by external reasons unrelated to subsidiary bank risk-taking, mitigating concerns about endogeneity. This allows us to establish a causal relationship between parent

debt and subsidiary bank risk-taking. We implement a difference-in-differences (DID) analysis, defining treatment and control groups. The control group consists of BHCs with a non-bank asset ratio of zero, while the treatment group includes BHCs with a non-bank asset ratio above the 90th percentile threshold before the pandemic. Our DID results confirm that, following the COVID-19 outbreak, BHCs with high non-bank asset ratios significantly reduced subsidiary banks' risk-taking. To further sharpen identification, we conduct a subsample analysis around the COVID-19 outbreak, leveraging the sharp increase in parent debt issuance during this period. Our results confirm that a higher parent debt ratio reduces subsidiary bank risk-taking, reinforcing the robustness of our findings and mitigating concerns about sample selection bias.

We present a simple model that captures our empirical finding. In the model, the BHC comprises both a bank subsidiary and a non-bank subsidiary. The parent company issues debt and allocates the proceeds as equity investments into these subsidiaries. The bank subsidiary faces investment opportunities that involve a trade-off between risk and return. our model predicts that when the parent company issues more debt and invests the proceeds as equity investment into its subsidiaries, it can mitigate the subsidiary bank's risk-taking behavior. Conversely, if the parent company issues less debt, the subsidiary bank needs to increase its own debt issuance to finance its investments, which can incentivize greater risk-taking. This occurs because, under limited liability, the downside is shared with debt holders, while shareholders pocket the upside.

This study contributes to the large literature on the financial ratio regulations aimed at curbing financial institutions' risk-taking incentives. The specific ratio examined here is the parent company's capital structure, which differs from prior research that primarily focus on capital structure at the commercial bank level or the consolidated BHC level

(e.g., [Keeley \[1990\]](#), [Shrieves and Dahl \[1992\]](#), [Demsetz and Strahan \[1997\]](#), and [Ellul and Yerramilli \[2013\]](#)).

This paper also contributes to the literature on the organizational structure of financial institutions and risk-taking. Our analysis focuses on the multi-level organizational structures of BHCs and their distinct balance sheets—particularly that of the parent company—rather than the single balance sheet of integrated conglomerates emphasized by [Freixas et al. \[2007\]](#). Moreover, we highlight the importance of considering existing BHC organizational structures when assessing risk-taking, in contrast to [Kahn and Winton \[2004\]](#) and [Luciano and Wihlborg \[2018\]](#), who design optimal structures to mitigate risk-taking. Closest to our approach is [Li et al. \[2024\]](#), who argue that a bank’s organizational structure should be evaluated alongside its capital structure and bankers’ compensation contracts when discouraging banker’s risk-taking.

Finally, we add to the nascent literature on parent debt in BHCs, where existing research is scarce. [Tucker et al. \[2023\]](#) show that the parent-only leverage ratio is positively associated with the parent’s credit default swap premium, suggesting that parent-only balance sheet data can inform assessments of parent credit risk and that consolidated reporting can obscure critical entity-level details. Building on these insights, we examine how parent company debt affects subsidiary bank risk-taking, offering a novel perspective on risk transmission within BHCs.

The paper is structured as follows: Section 2 describes the data sources, key variables, and summary statistics. Section 3 conducts the empirical analysis on the relationship between parent debt and subsidiary banks’ risk-taking. Section 4 presents the difference-in-differences analysis. Section 5 examines the impact of the cross-guarantee mechanism. Section 6 conducts robustness test. Section 7 introduces the theoretical model. Finally, Section 8 concludes.

2 Data and summary statistics

In this section, we describe our data sample, outline the construction of the parent debt ratio and subsidiary banks' risk-taking measures, and summarize our sample. Table 1 provides detailed definition of these variables.

2.1 Data sources

2.1.1 BHC-level data

Our BHC-level data are sourced from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP) provided by the Federal Reserve Bank. We use data spanning from 2012 to 2022, which include quarterly observations of the income statement and balance sheets of all U.S. BHCs and their parent-only companies. We drop BHCs that do not file the FR-Y9C or FR-Y9LP for each quarter during the entire sample period to avoid the impact of BHCs' joining or exiting on our empirical results. Finally, we collect data for 123 BHCs over the sample period.

2.1.2 Bank-level data

The subsidiary bank-level data are from the U.S. Call Reports provided by the Federal Reserve Bank, covering the period from 2012 to 2022. This dataset contains quarterly observations of the income statement and balance sheets of all U.S. commercial banks, some of which are the subsidiaries of BHCs. The data contain bank-level identifier

(RSSD9348) that can be used to link to the BHC-level data. Finally, our balanced panel consists of 157 unique bank subsidiaries.

2.2 Parent debt ratio

Our key explanatory variable is the parent debt ratio, defined as the parent’s non-deposit liability scaled by the BHC’s non-deposit liability. The parent’s non-deposit liability includes the parent company’s securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures. BHC’s non-deposit liability contains group-level consolidated total liabilities but excludes deposits in domestic offices, foreign offices, Edge and Agreement subsidiaries, and international banking facilities (IBFs).

2.3 Subsidiary bank’s risk-taking

To better measure bank’s risk, we mainly focus on subsidiary bank’s credit risk from both the asset and liability side. We also use Z-Score to measure bank’s risk, defined as the return on assets plus equity-to-asset ratio, scaled by the standard deviation of return on assets, where the equity-to-asset ratio is calculated as (book equity)/(total assets), thus, Z-Score comprises risk measure from both the liability and asset sides. A higher Z-Score indicates greater bank stability. Due to the skewness of Z-Score, we use its natural logarithm, which follows a normal distribution ([Laeven and Levine \[2007\]](#)). Following [Li and Song \[2022\]](#), we also use the risk-weighted leverage, defined as a bank’s risk-weighted asset divided by tier-1 capital as our risk-taking measure. The risk-weighted leverage is the product of a bank’s asset risk weight and leverage and thus summarizes a bank’s risk-

taking on both margins, therefore, we also include both asset risk weight and leverage as our dependent variables. We winsorize all the variables at the 1% level to minimize the impact of outliers.

2.4 Control variables

We use market data from the Center for Research in Security Prices (CRSP) and regulatory accounting data from Call Reports, FR Y-9C, and FR Y-9LP to construct bank- and BHC-level controls that are considered as important determinants of a bank’s risk-taking behavior.

These control variables include bank size ($LnTA$), leverage ratio ($Debt/Equity$), and Tobin’s q of the BHCs ($TobinQ$). [Keeley \[1990\]](#) shows that Tobin Q , as a proxy of market power or charter value, is negatively related to a bank’s risk-taking. [Boyd and Runkle \[1993\]](#) suggest a negative relationship between bank size and bank risk. [Acosta-Smith et al. \[2020\]](#) find that a leverage ratio requirement associated with the introduction of the Basel III Leverage Ratio can incentivize banks bound by it to increase their risk-taking.

2.5 Summary statistics

Table 2, Panel A, summarizes the sample statistics for the 2012-2022 period. The parent debt ratio averages 11.10%, ranging from 0 to 75.6%. The mean logarithm of asset size is 15.53, which translates to a mean of USD 5.53 billion in total assets. The average Tobin’s Q for BHCs is 1.04.

Regarding subsidiary banks’ risk-taking measures, the mean logarithm of the Z-score is 3.55, the average risk-weighted leverage is 6.86, the risk weight is 0.67, and the leverage ratio is 9.84.

3 Empirical approach and results

The central premise of our paper is the impact of parent debt ratio on the risk-taking of subsidiary banks. In this section, we present evidence that establishes the negative link between the parent debt ratio and subsidiary bank’s risk-taking.

For each subsidiary bank, we estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = & \alpha + \beta ParentDebtRatio_{j,t-4} + \\ & + \gamma Controls_{i,t-4} + \delta TobinQ_{j,t-4} + \lambda_i + \eta_t + \epsilon_{ijt} \end{aligned} \quad (1)$$

where, for each subsidiary bank i affiliated with BHC j and quarter t , *ParentDebtRatio* is calculated as the parent’s non-deposit liability divided by the BHC’s non-deposit liability, *Controls* represents a set of bank-level control variables described in Section 2.4. *TobinQ* is a BHC-level variable. Since most subsidiary banks are not public, we could not calculate the bank-level Tobin Q. Therefore, we use BHC-level Tobin Q as an alternative in our regression model. All explanatory variables are lagged by four quarters to mitigate concerns over look-ahead bias. *RiskTaking* is a measure of bank’s risk-taking originating in quarter t described in Section 2.3. We include subsidiary bank and quarter fixed effects to mitigate omitted-variable bias. In all models, standard errors are clustered at subsidiary bank level.

Table 3 presents the empirical evidence showing a negative relationship between the parent debt ratio and the subsidiary bank’s risk-taking during the 2012–2022 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.129), we observe an expected 4.86% increase in Z-Score ($0.3766 \times 0.129 \times 100\% = 4.86\%$), a 15.90% decrease in risk-weighted leverage ($-1.2323 \times 0.129 \times 100\% = -15.90\%$), and a 20.23% decrease in leverage ($-1.5685 \times 0.129 \times 100\% = -20.23\%$).

Overall, the empirical evidence suggests that parent debt reduces subsidiary banks’ risk-taking.

4 Identification strategy: difference-in-differences

The empirical results in Section 3 show a negative relationship between parent debt and subsidiary banks’ risk-taking, but this finding may be confounded by omitted variable bias. For instance, when a subsidiary bank faces stringent regulatory capital requirements, it may reduce its risk-taking to avoid potential capital shortfalls. At the same time, the parent company may issue additional external debt and inject the proceeds into the subsidiary bank to help it meet these requirements. In this case, the observed negative relationship is likely driven by the omitted factor—namely, the tightening of regulatory capital constraints—rather than a direct causal effect of parent debt.

To establish a causal link between parent debt ratio and the subsidiary bank’s risk-taking, we employ an instrumental variable (IV) that creates plausibly exogenous variation in parent debt ratio. Parent debt ratio are more likely to be affected by the breakout of COVID-19. Therefore, we instrument the BHC’s parent debt ratio with the timing of COVID-19 outbreak. COVID-19 began impacting the U.S. BHCs from the second

quarter of 2020, inducing a time-serious variation in parent debt ratio during our sample period. Moreover, the outbreak of COVID-19 is an exogenous event, unlikely to be affected by the BHC’s financing policy. A critical assumption of this IV approach is that COVID-19 affects subsidiary bank’s risk-taking only through its impact on parent debt ratio and not through other mechanisms. Since COVID-19 has impacted all BHCs in the U.S., to disentangle the different effects of parent debt on bank subsidiary’s risk-taking across BHCs, we implement a difference-in-differences (DID) analysis by defining treatment and control groups. BHCs typically have both bank and non-bank subsidiaries. If the proportion of non-bank assets within the BHC is high, the parent may have greater incentive to reduce subsidiary bank’s risk-taking when issuing parent debt. This is because, if subsidiary bank takes excessive risk and fails, the parent may need to utilize the non-bank assets to bail in subsidiary bank, potentially affecting the non-bank’s profit since the non-bank’s profit is usually higher (Lu et al. [2023]). Based on this rationale, the control group includes BHCs without any non-bank assets during our sample period, while the treatment group contains BHCs, whose non-bank asset ratio is above the 90th percentile threshold of all BHCs prior to the COVID-19 pandemic.

The difference-in-differences specification is as follows:

$$RiskTaking_{ijt} = \alpha + \beta_1 Treated + \beta_2 Post + \beta_3 Treated * Post + \gamma Controls_{i,t-4} + \lambda_i + \epsilon_{ijt}, \quad (2)$$

where i , j and t denote the subsidiary bank, BHC, and quarter, respectively. The variable $Post$ equals one for the post-COVID period and zero for the pre-COVID period. $Treated$ is equal to one for the treated sample and zero for the control sample. $Controls$ represent a set of subsidiary bank- and BHC-level control variables described in Section

2.4. We include subsidiary bank fixed effects. We focus on the difference-in-differences estimator, β_3 , which is the coefficient on the product of the *Treated* and *Post*.

[Bertrand et al. \[2004\]](#) argue that the standard errors of a pooled difference-in-differences estimator are generally underestimated. To address this concern, they recommend aggregating each firm’s pre- and post-treatment data into a single pre- and single post-observation. Following [Bertrand et al. \[2004\]](#) and [Pogach and Unal \[2019\]](#), we construct risk-taking measures for each subsidiary bank in the pre- and post-COVID periods.

Table 2, Panel B, presents the statistics of the reduced sample from 2017 to 2022. The parent debt ratio averages 17.00%, while the mean logarithm of asset size is 15.98. The average Tobin’s Q for BHCs is 1.04. Compared with the whole sample, both the mean parent debt ratio and asset size increased slightly during this period.

Table 4 presents the regression results for the difference-in-differences analysis. Model (1) shows that the coefficient of interest on $Treated \times Post$ is positive and significant at the 5% level. Specifically, the logarithm of Z-Score for the treated group increases by 33.84 log points in the post-COVID period relative to the control group, suggesting that a higher parent debt ratio reduces subsidiary banks’ overall risk. The effect is still robust after including subsidiary bank- and BHC-level controls. Model (6) shows that the coefficient of interest on $Treated \times Post$ is negative and significant at the 1% level, suggesting that the risk weight for the treated group decreases by 4.98 percentage points in the post-COVID period relative to the control group, which further confirms that a higher parent debt ratio reduces subsidiary banks’ overall risk.

Overall, the difference-in-differences analysis reported in this section indicates a causal effect of parent debt on subsidiary banks’ risk-taking; that is, a higher parent debt ratio reduces subsidiary banks’ risk-taking.

5 The impact of cross-guarantee on subsidiary bank’s risk-taking

Our previous analysis demonstrates that parent debt reduces risk-taking at the subsidiary bank level. We propose two complementary mechanisms to explain this finding.

The first mechanism operates through joint liability. When a parent holding company defaults, its subsidiary banks may bear liability for the parent’s obligations, effectively creating a contractual bail-in arrangement. This relationship is reciprocal: subsidiary bank-level distress can similarly propagate upward to the parent.

The second mechanism stems from the cross-guarantee provisions of the Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA). Under FIRREA, the Federal Deposit Insurance Corporation (FDIC) is permitted to charge off any expected losses related to the failure of one subsidiary bank within a multi-bank holding company to the capital of affiliated subsidiary banks within the same organization ([Ashcraft \[2005\]](#)). We illustrate this with an example: consider two BHCs — A and B. BHC A operates three bank subsidiaries (A1, A2, and A3), whereas BHC B operates a single subsidiary (B1). If A1 defaults, the FDIC may require BHC A to deploy capital from A2 and A3 to absorb A1’s losses before any public intervention occurs. By contrast, if B1 defaults, BHC B has no affiliated subsidiaries from which to draw capital, necessitating an FDIC bailout. This asymmetry suggests that multi-bank BHCs face

stronger internal discipline: because they must deploy their own capital to resolve a failing subsidiary, they have greater incentive to limit risk-taking ex ante.

Building on this, we empirically test the regulatory bail-in mechanism by examining whether BHCs with multiple bank subsidiaries curtail subsidiary banks' risk-taking when the parent issues more external debt. For each bank subsidiary, we estimate the following model:

$$\begin{aligned}
RiskTaking_{ijt} = & \alpha + \beta_1 ParentDebtRatio_{j,t-4} + \beta_2 Multibank_{j,t} + \\
& + \beta_3 Multibank_{j,t} * ParentDebtRatio_{j,t-4} + \\
& + \gamma Control_{i,t-4} + \delta TobinQ_{j,t-4} + \lambda_i + \eta_t + \epsilon_{ijt}
\end{aligned} \tag{3}$$

where, for each bank subsidiary i affiliated with BHC j and in quarter t , *Multibank* equals to one if the BHC has more than one subsidiary banks, otherwise, zero. *Control* and *TobinQ* are the same as these we used in regression model (1). Table 2, Panel C, summarizes key variables used in the model.

Table 5 demonstrates a negative relationship between the cross-guarantee and subsidiary bank's risk-taking. For a one standard deviation increase in parent debt ratio (0.129), we observe an expected additional 21.66% decrease in the risk-weighted leverage ($0.129 \times -1.6793 \times 100\% = -21.66\%$) and a 2.53% decrease in the risk weight ($0.129 \times -0.1958 \times 100\% = -2.52\%$) if the BHC has more than one subsidiary banks.

Overall, the results presented in this section show that parent debt reduces subsidiary banks' risk-taking especially when there is more than one subsidiary bank within the BHC, potentially due to the cross-guarantee mechanism, which gives the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses.

6 Robustness test

The robustness check addresses concern regarding whether our results are truly driven by sample selection.

Table 2, Panel D, summarizes the sample statistics for the 2017–2022 period. On average, the parent debt ratio is 13.7%. The mean logarithm of asset size is 15.76. The average debt/equity ratio is 7.24, while the mean Tobin’s Q of BHCs is 1.04.

In the difference-in-differences analysis, we use a sample from 2017 to 2022. For the robustness check, we limit our sample to this same period, that is, 2017–2022. Table 6 shows a negative relationship between the parent debt ratio and the subsidiary bank’s risk-taking. Specifically, for a one standard deviation increase in the parent debt ratio (0.139), we observe an expected 5.08% increase in Z-Score ($0.3655 \times 0.139 \times 100\% = 5.08\%$), a 7.03% decrease in risk-weighted leverage ($-0.5062 \times 0.139 \times 100\% = -7.03\%$), and a 23.85% decrease in leverage ($-1.7158 \times 0.139 \times 100\% = -23.85\%$).

The robustness test confirms that a higher parent debt ratio reduces subsidiary banks’ risk-taking and that the main result is not driven by the sample selection.

7 A model of parent debt and subsidiary banks’ risk-taking

This section develops a theoretical framework to analyze the relationship between parent debt and the risk-taking behavior of subsidiary banks. Detailed derivations and proofs are provided in the Appendix. The model operates over two periods, denoted as $t =$

0 and $t = 1$. We consider a BHC that wholly owns two distinct entities within its organizational structure: a subsidiary bank and a nonbank subsidiary.

7.1 Period $t = 0$

At $t = 0$, the consolidated balance sheet of the BHC comprises one unit of asset, D_c units of debt, and E_c units of equity. In addition to the consolidated balance sheet, the BHC maintains a parent-company-only balance sheet, which is subject to Federal Reserve oversight. The parent company holds E_p units of equity and issues D_p units of debt. It downstreams these funds to capitalize both the subsidiary bank and non-bank subsidiary proportionally. Specifically, the parent injects $\alpha(E_p + D_p)$ into the nonbank subsidiary and $(1 - \alpha)(E_p + D_p)$ into the subsidiary bank as equity investments, where α and $1 - \alpha$ represent the relative asset sizes of the nonbank subsidiary and subsidiary bank, respectively.

Under this formulation,

$$E_p + D_p \tag{1}$$

represent the uniform equity capital ratio for both subsidiaries, as capitalized by the parent company. While capital allocation within a BHC is typically endogenous, we treat the subsidiary capital ratios as fixed parameters. This simplifying assumption can be motivated by the presence of binding regulatory capital requirements at both the subsidiary bank and the BHC levels. The nonbank subsidiary's assets, debt, and equity are denoted by α , D_n , and E_n , respectively. Similarly, the subsidiary bank's assets, debt, and equity are denoted by $1 - \alpha$, D_b , and E_b . The balance sheets for the consolidated BHC, the parent company, the nonbank subsidiary, and the subsidiary

bank are summarized in Table 1. By construction, E_b and E_n constitute the assets of the parent company (A_b and A_n).

Table 1: Balance Sheets at $t = 0$

Consolidated BHC	Parent Company	Nonbank Subsidiary	Subsidiary Bank
$\begin{array}{c c} 1 & D_c \\ \hline & E_c \end{array}$	$\begin{array}{c c} A_b & E_p \\ \hline A_n & D_p \end{array}$	$\begin{array}{c c} \alpha & E_n \\ \hline & D_n \end{array}$	$\begin{array}{c c} 1 - \alpha & E_b \\ \hline & D_b \end{array}$

At $t = 0$, the subsidiary bank encounters an investment opportunity that yields a payoff R_b with probability $P(R_b)$ and a salvage value S with probability $1 - P(R_b)$. To capture a classical risk-return trade-off, we assume the probability of success is a decreasing, linear function of the target payoff:

$$P(R_b) = 1 - \beta R_b, \quad (2)$$

implying that higher returns are inherently less likely. The payoff distribution of the subsidiary bank's investment is therefore:

$$R = \begin{cases} R_b, & \text{with probability } 1 - \beta R_b, \\ S, & \text{with probability } \beta R_b. \end{cases}$$

To ensure that the limited liability friction is economically meaningful, we impose the following condition on the magnitude of the salvage value:

$$S < 1 - E_p. \quad (3)$$

This condition guarantees that, in the absence of capitalization financed by parent debt ($D_p = 0$), the potential loss from the investment is sufficiently severe to render the subsidiary bank insolvent; that is, $S(1 - \alpha) < (1 - \alpha)(1 - E_p)$. If we were to assume the

alternative ($S \geq 1 - E_p$), the equity capital endogenously supplied by the BHC would always be abundant enough to absorb the investment losses, preventing subsidiary bank bankruptcy and neutralizing the core risk-shifting incentives.

The nonbank subsidiary generates a return of R_n . For simplicity, we assume the nonbank activities yield a risk-free return, denoted as r_f . This assumption reflects the reality that many nonbank operations—such as advisory services, asset management, and securities underwriting—generate stable fee income that lacks the default risk typical of bank lending portfolios. Normalizing the risk-free rate to 1, and assuming the subsidiary bank does not exercise deposit market power in its deposit rate setting, we establish:

$$r_b = r_f = 1. \quad (4)$$

This provides a neutral benchmark where the subsidiary bank cannot extract rents on the liability side.

7.2 Period $t = 1$

At $t = 1$, the realized balance sheets for the subsidiary bank are determined by the outcome of the subsidiary bank's investment. The balance sheets for the nonbank subsidiary, and the subsidiary bank are summarized in Table 2.

Table 2: Subsidiaries' Balance Sheets at $t = 1$

Nonbank Subsidiary	Subsidiary Bank (Success)	Subsidiary Bank
$R_n\alpha \mid \begin{array}{c} R_n\alpha - D_n \\ D_n \end{array}$	$R_b(1 - \alpha) \mid \begin{array}{c} R_b(1 - \alpha) - D_b \\ D_b \end{array}$	$S(1 - \alpha) \mid \begin{array}{c} \max\{S(1 - \alpha) - D_b, 0\} \\ D_b \end{array}$

The consolidated BHC shareholder chooses the risk profile R_b to maximize the expected payoff. The objective function is given by:

$$\begin{aligned} \max_{R_b} \quad \mathbb{E}[L] = & P(R_b) [R_n \alpha - D_n + R_b(1 - \alpha) - D_b - D_p] \\ & + (1 - P(R_b)) [R_n \alpha - D_n + \max\{S(1 - \alpha) - D_b, 0\} - D_p], \end{aligned} \quad (5)$$

where the debt obligations of the two subsidiaries are strictly determined by the uniform parent equity allocation:

$$D_b = (1 - \alpha)(1 - E_p - D_p), \quad (6)$$

$$D_n = \alpha(1 - E_p - D_p). \quad (7)$$

Before analyzing the equilibrium under the limited liability distortion, it is instructive to establish the first-best investment policy. The first-best risk level is determined by maximizing the expected investment payoff of the subsidiary bank's production technology, without regard to the BHC's capital structure or moral hazard. Taking the first-order derivative of the expected payoff with respect to R_b yields the first-best solution:

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \quad (8)$$

The subsidiary bank fails if and only if its equity value falls below its debt obligation, $S(1 - \alpha) < D_b$. Substituting Equation (6), this condition simplifies to:

$$D_p < (1 - E_p) - S \equiv \hat{D}'_p. \quad (9)$$

Furthermore, when the subsidiary bank fails, the parent/BHC survives if and only if the residual equity value of the nonbank subsidiary is sufficient to cover the parent's independent debt obligation D_p . This survival condition is $R_n\alpha - D_n - D_p \geq 0$, which, applying Equation (7), reduces to:

$$D_p \leq \frac{\alpha[R_n - (1 - E_p)]}{1 - \alpha} \equiv \hat{D}_p'' \quad (10)$$

Building on these thresholds, the subsequent analysis examines the BHC's optimal risk-taking decision across the two broad cases (1) no parent/BHC bankruptcy when the subsidiary bank fails, and (2) parent/BHC fails. When the parent/BHC fails, the subsidiary bank can fail or not.

7.3 Case 1: No parent/BHC bankruptcy ($D_p \leq \hat{D}_p''$)

We begin our equilibrium analysis with the parameter space where $D_p \leq \hat{D}_p''$. In this regime, the BHC possesses sufficient nonbank equity to fully absorb the parent debt obligation, ensuring it survives even if the subsidiary bank fails on its investment.

Lemma 1. *If and only if $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \hat{\alpha}$, it holds that $\hat{D}_p' < \hat{D}_p''$. Excessive risk-taking occurs when $D_p < \hat{D}_p'$, whereas the first-best risk level emerges when $D_p \in [\hat{D}_p', \hat{D}_p'']$.*

See Appendix A1: Proof of Lemma 1.

When $D_p < \hat{D}_p'$, the subsidiary bank fails under distress (realizing salvage value S), but the BHC survives. Limited liability protection at the subsidiary bank level distorts

the risk-taking incentives of the BHC shareholder. Mathematically, solving the FOC of the objective function yields the optimal risk level:

$$R_b^* = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}. \quad (11)$$

Because the default condition ensures $1 - E_p - D_p > S$, it strictly follows that $R_b^* > R_b^{FB}$, confirming that excessive risk-taking occurs.

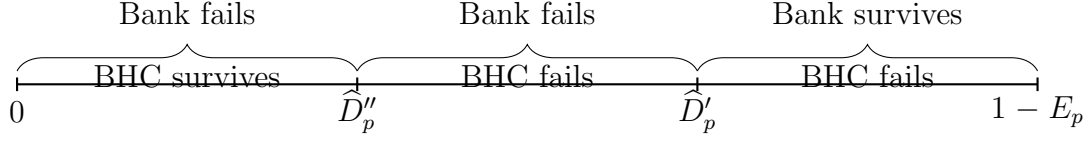
In contrast, when $D_p \in [\hat{D}_p', \hat{D}_p'']$, both the subsidiary bank and the parent/BHC survive the bad state. The losses are fully passed onto the shareholder of the BHC. The absence of limited liability distortion on the subsidiary bank level fully aligns the private risk-taking decision with the first-best level, yielding the first-best outcome, R_b^{FB} .

Lemma 2. *A more significant nonbank subsidiary, i.e., an increase in α (given $\alpha > \hat{\alpha}$), reduces risk-taking for a given D_p .*

An increase in α acts to increase the parent survival threshold $\hat{D}_p''$, while the subsidiary bank failure threshold $\hat{D}_p'$ remains unchanged. This cleanly expands the parameter interval $[\hat{D}_p', \hat{D}_p'']$ governed by the first-best risk level. Intuitively, when α approaches 0 and the equity value from the nonbank subsidiary becomes negligible, the limited liability distortion is fully transmitted onto the parent/BHC level, creating excessive risk-taking incentives.

Lemma 3. *If and only if $\alpha < \frac{(1-E_p)-S}{R_n-S} \equiv \hat{\alpha}$, it holds that $\hat{D}_p' > \hat{D}_p''$. Excessive risk-taking occurs when $D_p < \hat{D}_p''$ if the parent/BHC does not fail.*

When α falls below $\hat{\alpha}$, the ordering of the critical thresholds flips. The possible outcome boundaries are visually summarized in the figure below:



7.4 Case 2: Parent/BHC bankruptcy risk ($D_p > \hat{D}_p''$)

We now shift to the parameter space where $D_p > \hat{D}_p''$, exposing the parent company to severe bankruptcy risk. When the parent/BHC fails, its shareholders receive a terminal equity payoff of exactly zero, tightly shielded by parent-level limited liability.

7.4.1 Both the subsidiary bank and the parent/BHC fail ($\hat{D}_p'' < D_p < \hat{D}_p'$)

Lemma 4. *When both the parent/BHC and subsidiary bank fail, excessive risk-taking still occurs because limited liability distortion emerges at the subsidiary bank level.*

See Appendix A2: Proof of Lemma 4.

Mathematically, the optimal risk level chosen by the BHC shareholder in this regime is given by:

$$R_b = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}. \quad (12)$$

The subsidiary bank still takes excessive risk because the limited liability constraint binds strictly at the subsidiary bank level.

7.4.2 The parent/BHC fails, but the subsidiary bank survives ($D_p > \hat{D}'_p$)

Lemma 5. *When the parent/BHC fails but the subsidiary bank remains solvent, it yields the first best solution with*

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \quad (13)$$

See Appendix A3: Proof of Lemma 5.

When the parent/BHC fails but the subsidiary bank remains solvent, the absence of a limited liability distortion at the subsidiary bank level aligns the private risk-taking decision directly with the socially optimal first-best level.

8 Conclusion

This paper provides evidence on how parent debt affects subsidiary banks' risk-taking. Using a comprehensive dataset covering all domestic U.S. commercial banks and their corresponding BHCs from 2012 to 2022, combined with a difference-in-differences analysis using the COVID-19 outbreak as a natural experiment, we find that increased parent debt is associated with reduced risk-taking by subsidiary banks. This effect is particularly prominent in BHCs with significant nonbank business or more than one bank subsidiary, which would give the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

References

- Jonathan Acosta-Smith, Michael Grill, and Jan Hannes Lang. The leverage ratio, risk-taking and bank stability. *Journal of Financial Stability*, page 100833, 2020.
- Adam B Ashcraft. Are banks really special? new evidence from the fdic-induced failure of healthy banks. *American Economic Review*, 95(5):1712–1730, 2005.
- Dafna Avraham, Patricia Selvaggi, and James I Vickery. A structural view of us bank holding companies. *Economic Policy Review*, 18(2):65–81, 2012.
- Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275, 2004.
- John H Boyd and David E Runkle. Size and performance of banking firms: Testing the predictions of theory. *Journal of Monetary Economics*, 31(1):47–67, 1993.
- Nicola Cetorelli and Linda S Goldberg. Banking globalization and monetary transmission. *The Journal of Finance*, 67(5):1811–1843, 2012.
- Rebecca S Demsetz and Philip E Strahan. Diversification, size, and risk at bank holding companies. *Journal of Money, Credit, and Banking*, pages 300–313, 1997.
- Andrew Ellul and Vijay Yerramilli. Stronger risk controls, lower risk: Evidence from us bank holding companies. *The Journal of Finance*, 68(5):1757–1803, 2013.
- Xavier Freixas, Gyöngyi Lóránth, and Alan D Morrison. Regulating financial conglomerates. *Journal of Financial Intermediation*, 16(4):479–514, 2007.

- Michael C Jensen and William H Meckling. Theory of the firm. *Managerial behavior, agency costs and ownership structure*, 3(4):305–360, 1976.
- Charles Kahn and Andrew Winton. Moral hazard and optimal subsidiary structure for financial institutions. *The Journal of Finance*, 59(6):2531–2575, 2004.
- Michael C Keeley. Deposit insurance, risk, and market power in banking. *The American Economic Review*, pages 1183–1200, 1990.
- Luc Laeven and Ross Levine. Is there a diversification discount in financial conglomerates? *Journal of Financial Economics*, 85(2):331–367, 2007.
- Zhao Li, Kebin Ma, and Lucy White. The making of (modern) banks. *WBS Finance Group Research Paper*, 2024.
- Ziang Li and Jihong Song. Deposit market power and bank risk-taking. 2022.
- Lina Lu, Marco Macchiavelli, and Jonathan Wallen. Internal and external capital markets of large banks. *Available at SSRN*, 2023.
- Elisa Luciano and Clas Wihlborg. Financial synergies and systemic risk in the organization of bank affiliates. *Journal of Banking & Finance*, 88:208–224, 2018.
- Jonathan Pogach and Haluk Unal. The dark-side of banks’ nonbank business: Internal dividends in bank holding companies. *FDIC Center for Financial Research Paper*, (2018-01), 2019.
- Ronald E Shrieves and Drew Dahl. The relationship between risk and capital in commercial banks. *Journal of Banking & Finance*, 16(2):439–457, 1992.
- Jennifer W Tucker, Ying Zhou, and Jigao Zhu. Parent-only balance sheet information and credit risk assessments. *Accounting Horizons*, pages 1–22, 2023.

Table 1

Variable definitions and sources of data used in the study. This table summarizes the variable definitions and sources of data used in our study.

Variables	Definition
Tobin Q	Tobin Q is the group-level consolidated tobin's q, and defined as the ratio between the market value of BHC (MV) over the replacement cost of its assets (RC). It is estimated by $Tobin\ Q = (AT + ME - BE) / AT$, where AT is total asset, ME is the market value of equity at year end by multiplying stock price with the common shares outstanding, BE is (Stockholders Equity + Deferred Taxes + Investment Tax Credit - Preferred Stock). I use the code from WRDS to calculate it. Please see the link for details: https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/
Z-Score	Z-score is a proxy for bank stability. It is estimated as return on assets plus equity-to-asset ratio, scaled by the standard deviation of return on assets, where equity-to-asset ratio is estimated by (book equity)/(total assets). The variable is calculated at the annual level. To avoid the skewness, we take the logarithm of it.
ROA	ROA is estimated as net income divided by total assets: RIAD4340(bank subsidiary's net income (loss) attributable to bank from FFIEC041)/ RCON2170(bank subsidiary's total asset from FFIEC 041) or RIAD4340(bank subsidiary's net income (loss) attributable to bank from FFIEC031)/RCFD2170 (bank subsidiary's total asset from FFIEC 031)
SD ROA	Standard deviation of subsidiary bank 's return on asset is a proxy for bank riskiness behavior. It is estimated as the standard deviation of the quarterly return on assets in a given year. The variable is calculated at the annual level.
Debt/Equity	Debt/Equity is the ratio of commercial bank's total liability divided by its total equity: $RCON2948(\text{bank subsidiary's total liability from FFIEC 041}) / (RCON2170(\text{bank subsidiary's total asset from FFIEC 041}) - RCON2948(\text{bank subsidiary's total liability from FFIEC 041}))$ or $RCFD2948(\text{bank subsidiary's total liability from FFIEC 031}) / (RCFD2170(\text{bank subsidiary's total asset from FFIEC 031}) - RCFD2948(\text{bank subsidiary's total liability from FFIEC 031}))$
Ln AT	Ln AT is the logarithm of subsidiary bank 's total assets: $\ln(RCON2170(\text{bank subsidiary's total asset from FFIEC 041}))$ or $\ln(RCFD2170((\text{bank subsidiary's total asset from FFIEC 031})))$

Parent Debt Ratio	Parent Debt Ratio is the ratio of the parent company's non-deposit debt liabilities to BHC's non-deposit liability: quarterly (BHCP2309(parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332(parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368(parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062(parent company's subordinated notes and debentures)+ BHCP0279(parent company's securities sold under agreements to repurchase))/(BHCK2948(group-level consolidated total liabilities)-BHDM6631(group-level consolidated noninterest-bearing deposits in domestic offices)-BHDM6636(group-level consolidated interest-bearing deposits in domestic offices)-BHFN6631(group-level consolidated noninterest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs)-BHFN6636(group-level consolidated interest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs))
Multi-bank	If the BHC has more than one subsidiary banks, this value equals to one, otherwise, zero.
RW Assets/T1 Capital	Risk-weighted leverage is the ratio of risk-weighted assets to tier 1 capital. It is measured by the RCFAA223(Total risk-weighted assets)/RCFA8274(Tier 1 capital) for banks that file FFIEC031 or the RCOAA223(Total risk-weighted assets)/RCOA8274(Tier 1 capital) for banks that file FFIEC041.
RW Assets/Assets	Asset risk weight is measured by the RCFAA223 (Total risk-weighted assets)/RCFD2170(Total asset) for banks that file FFIEC031 or the RCOAA223(Total risk-weighted assets)/RCON2170(Total assets) for banks that file FFIEC041.
Assets /T1 Capital	Leverage is measured by RCFD2170(Total assets)/RCFA8274(Tier 1 capital) for banks that file FFIEC031 or RCON2170(Total assets)/RCOA8274(Tier 1 capital) for banks that file FFIEC041.
Non-bank Asset Ratio	Non-bank asset ratio is measured by BHCP4778(Total combined nonbank assets of nonbank subsidiaries))/the sum of RCFD2170(Total assets) of all bank subsidiaries within the BHC for banks that file FFIEC031 or BHCP4778(Total combined nonbank assets of nonbank subsidiaries))/the sum of RCON2170(Total assets) of all bank subsidiaries within the BHC that file FFIEC041.

Table 2

Summary statistics of key characteristics. This table summarizes the sample statistics of key subsidiary bank-level and BHC-level characteristics used in our study. Panel A uses the full sample of subsidiary banks and BHCs during the 2012-2022 period. Panel B uses the reduced sample for difference-in-differences models over the 2017-2022. Both the full and reduced samples comprise 123 unique BHCs and 157 subsidiary banks.

Panel A: Full Sample (2012-2022)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Parent Debt Ratio	6,904	0.111	0.129	0	0	0.0608	0.211	0.299	0	0.756
ROA	6,908	0.0157	0.0477	0.00235	0.00378	0.00706	0.0108	0.0167	-0.0878	0.888
Debt/Equity	6,908	7.243	3.259	0.298	6.154	7.588	9.163	10.36	0	32.71
Ln AT	6,908	15.53	2.354	12.89	14.51	15.60	16.87	18.50	8.009	21.97
Tobin Q	6,889	1.035	0.0629	0.979	1.000	1.023	1.055	1.094	0.913	1.606
Z-Score	6,908	3.546	0.555	3.033	3.291	3.513	3.736	4.073	1.978	5.824
RW Assets/T1 Capital	6,882	6.862	2.745	0.850	6.304	7.739	8.590	9.348	0.201	10.66
RW Assets/Assets	6,882	0.667	0.186	0.352	0.600	0.711	0.791	0.856	0.193	0.982
Assets /T1 Capital	6,908	9.840	3.542	2.155	9.137	10.63	11.89	13.09	1.002	16.46

Panel B: Sample for difference-in-differences analysis (2017-2022)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Parent Debt Ratio	1,272	0.170	0.161	0	0	0.137	0.296	0.366	0	0.756
ROA	1,272	0.0126	0.0273	0.00255	0.00405	0.00732	0.0111	0.0169	-0.0530	0.399
Debt/Equity	1,272	7.347	3.201	0.647	6.228	7.697	9.256	10.45	0.00206	17.53
Ln AT	1,272	15.98	2.100	14.07	15.03	15.77	16.79	18.66	10.84	21.97
Tobin Q	1,272	1.037	0.0701	0.979	0.998	1.024	1.053	1.100	0.914	1.606
Z-Score	1,272	3.526	0.593	2.969	3.264	3.483	3.709	4.084	1.978	5.824
SD-ROA	1,272	0.00685	0.0130	0.00241	0.00318	0.00393	0.00491	0.00914	0.00106	0.126
RW Assets/T1 Capital	1,260	6.713	2.662	0.772	6.420	7.632	8.279	8.977	0.201	10.66
RW Assets/Assets	1,260	0.649	0.199	0.296	0.576	0.696	0.791	0.852	0.193	0.982
Assets /T1 Capital	1,272	9.842	3.305	3.520	9.275	10.47	11.65	12.77	1.002	16.46
Nonbank	1,272	0.321	0.467	0	0	0	1	1	0	1
Time	1,272	0.458	0.498	0	0	0	1	1	0	1

Panel C: Sample for Cross-guarantee Analysis (2012-2022)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Parent Debt Ratio	6,904	0.111	0.129	0	0	0.0608	0.211	0.299	0	0.756
ROA	6,908	0.0157	0.0477	0.00235	0.00378	0.00706	0.0108	0.0167	-0.0878	0.888
Debt/Equity	6,908	7.243	3.259	0.298	6.154	7.588	9.163	10.36	0	32.71
Ln AT	6,908	15.53	2.354	12.89	14.51	15.60	16.87	18.50	8.009	21.97
Tobin Q	6,889	1.035	0.0629	0.979	1.000	1.023	1.055	1.094	0.913	1.606
Z-Score	6,908	3.546	0.555	3.033	3.291	3.513	3.736	4.073	1.978	5.824
RW Assets/T1 Capital	6,882	6.862	2.745	0.850	6.304	7.739	8.590	9.348	0.201	10.66
RW Assets/Assets	6,882	0.667	0.186	0.352	0.600	0.711	0.791	0.856	0.193	0.982
Assets /T1 Capital	6,908	9.840	3.542	2.155	9.137	10.63	11.89	13.09	1.002	16.46
Multi-bank	6,908	0.363	0.481	0	0	0	1	1	0	1

Panel D: Reduced Sample (2017-2022)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Parent Debt Ratio	3,764	0.137	0.139	0	0	0.103	0.239	0.330	0	0.756
ROA	3,768	0.0172	0.0485	0.00258	0.00406	0.00761	0.0115	0.0177	-0.0671	0.586
Debt/Equity	3,768	7.236	3.317	0.243	6.030	7.604	9.172	10.30	0	32.71
Ln AT	3,768	15.76	2.352	13.07	14.79	15.84	17.09	18.68	8.015	21.97
Tobin Q	3,767	1.039	0.0669	0.979	1.001	1.026	1.059	1.101	0.914	1.606
Z-Score	3,768	3.485	0.540	3.004	3.247	3.454	3.672	3.996	1.978	5.824
RW Assets/T1 Capital	3,742	6.817	2.730	0.748	6.358	7.755	8.501	9.144	0.201	10.66
RW Assets/Assets	3,742	0.668	0.187	0.349	0.606	0.721	0.791	0.850	0.193	0.982
Assets /T1 Capital	3,768	9.715	3.447	1.792	9.236	10.55	11.68	12.69	1.002	16.46

Table 3

Parent debt ratio and subsidiary bank's risk-taking. This table presents the results from the model for our preliminary analysis used in Section (4). The dependent variables are the logarithm of Z-score (Z-Score), Risk-weighted Leverage (RW Assets/T1 Capital), Asset Risk Weight (RW Assets/Assets) and Leverage (Assets /T1 Capital). The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), and (7) contain only parent debt ratio as a control, while Models (2), (4), (6), and (8) contain all set of controls. All models use quarterly data for 2012-2022. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

[illegible]

Table 4

Difference-in-Differences Analysis in Subsidiary Bank's Risk-taking. This table reports results from difference-in-differences regressions using Equation (2). The treated group includes subsidiary banks with the non-bank asset ratio greater than the 90th percentile before COVID-19, while the control group consists of subsidiary banks that do not have non-bank subsidiary during the whole sample period. Post is an indicator variable equal to zero before COVID-19 and one after. All columns collapse each pre- and post-COVID periods into a single observation for each subsidiary bank. Models (1), (3), (5), and (7) contain Nonbank and Time as controls, while Models (2), (4), (6), and (8) contain all set of controls. All models include subsidiary bank fixed effect. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

VARIABLES	(1) Z-Score	(2) Z-Score	(3) RW Assets/T1 Capital	(4) RW Assets/T1 Capital	(5) RW Assets/Assets	(6) RW Assets/Assets	(7) Assets /T1 Capital	(8) Assets /T1 Capital
Treated X Post	0.3884** (0.1544)	0.3374** (0.1301)	-0.0955 (0.1978)	-0.3499 (0.2149)	-0.0134 (0.0208)	-0.0498*** (0.0164)	0.2676 (0.3125)	0.4710*** (0.1388)
Post	-0.0893*** (0.0309)	0.0600 (0.1310)	-0.1090 (0.1077)	0.1277 (0.2170)	-0.0566*** (0.0092)	0.0324 (0.0200)	0.6372*** (0.1607)	-0.6781*** (0.1365)
Tobin Q		-3.2102** (1.5870)		6.2684 (3.9176)		0.8514*** (0.2997)		-7.3914*** (2.5486)
Ln AT		-0.5876 (0.3545)		-0.6331 (0.3795)		-0.1185*** (0.0343)		0.9709*** (0.3366)
Debt/Equity		-0.0649 (0.0635)		0.2012* (0.1156)		-0.0250*** (0.0088)		0.9033*** (0.0833)
Constant	3.5101*** (0.0264)	16.6485*** (5.8209)	6.7777*** (0.0453)	8.8657 (6.6099)	0.6777*** (0.0043)	1.8371*** (0.6469)	9.5104*** (0.0695)	-4.4056 (4.9362)
Observations	106	106	104	104	104	104	106	106
R-squared	0.8718	0.9059	0.9852	0.9877	0.9753	0.9856	0.9765	0.9956
Bank_Subsiary FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	NO	NO	NO	NO	NO	NO	NO	NO

Table 5

Cross-guarantee and subsidiary bank's risk-taking. This table reports results using Equation (3). Models (1), (3), (5), and (7) contain Multi-bank, Parent Debt Ratio, and the interaction item as controls, while Models (2), (4), (6), and (8) contain all set of controls. All models include subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

[illegible]

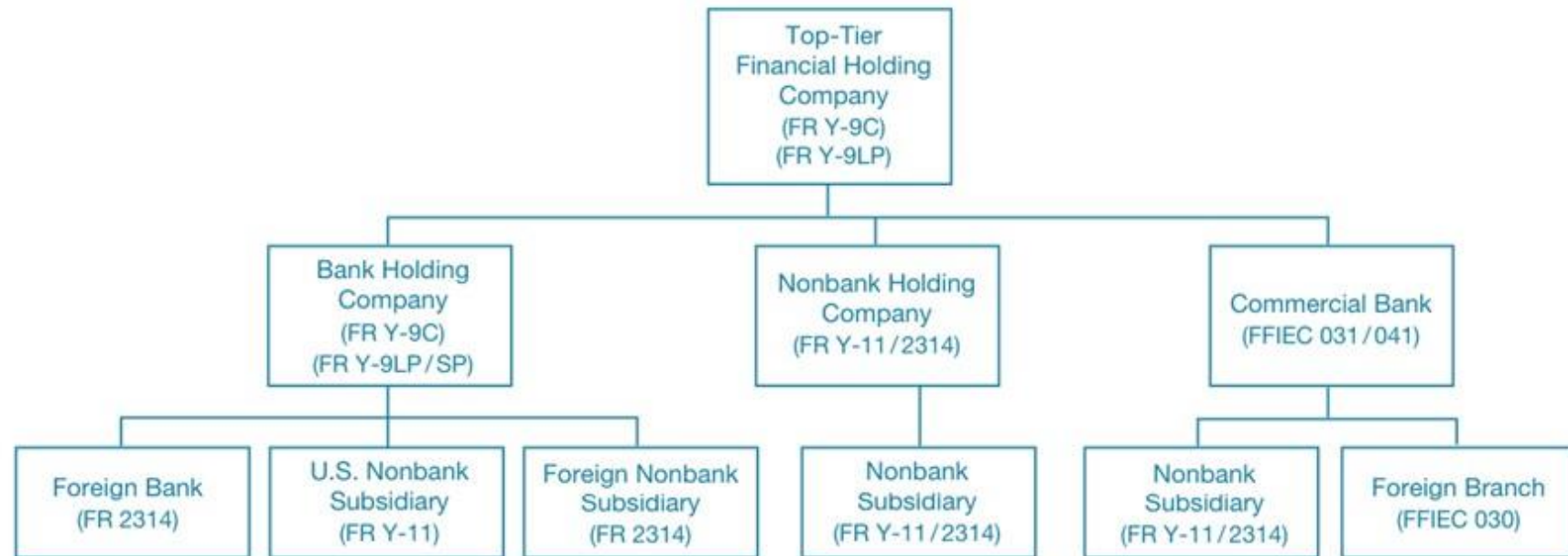
Table 6

Parent debt ratio and subsidiary bank's risk-taking. This table presents the results from the models for our preliminary analysis used in Section (7). The dependent variables are the logarithm of Z-score (Z-Score), Risk-weighted Leverage (RW Assets/T1 Capital), Asset Risk Weight (RW Assets/Assets) and Leverage (Assets /T1 Capital). The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), and (7) contain only parent debt ratio as a control, while Models (2), (4), (6), and (8) contain all set of controls. All models use quarterly data for 2017-2022. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

[illegible]

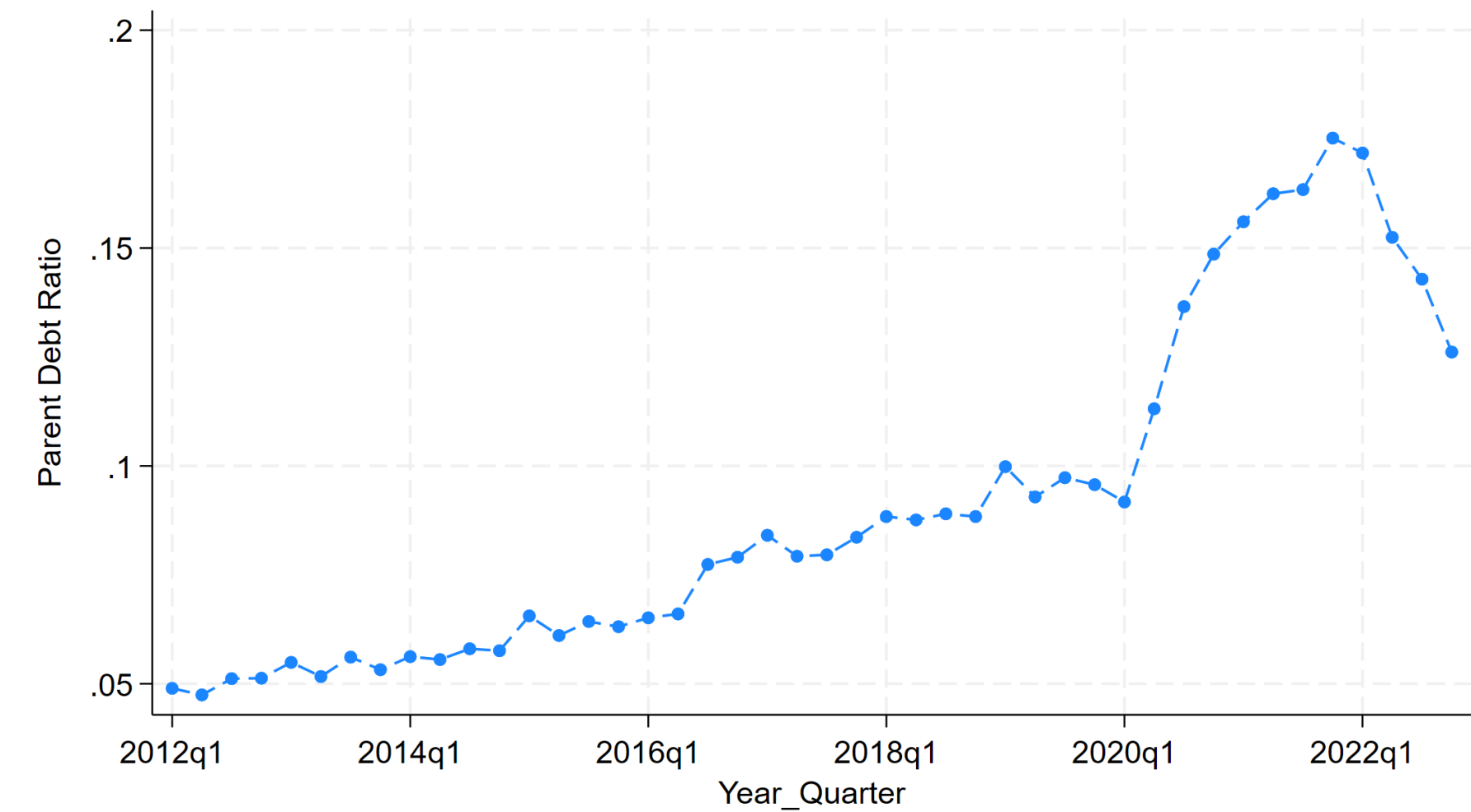
Figure 1: The organizational structure of a large BHC

Stylized Structure of a Large Bank Holding Company



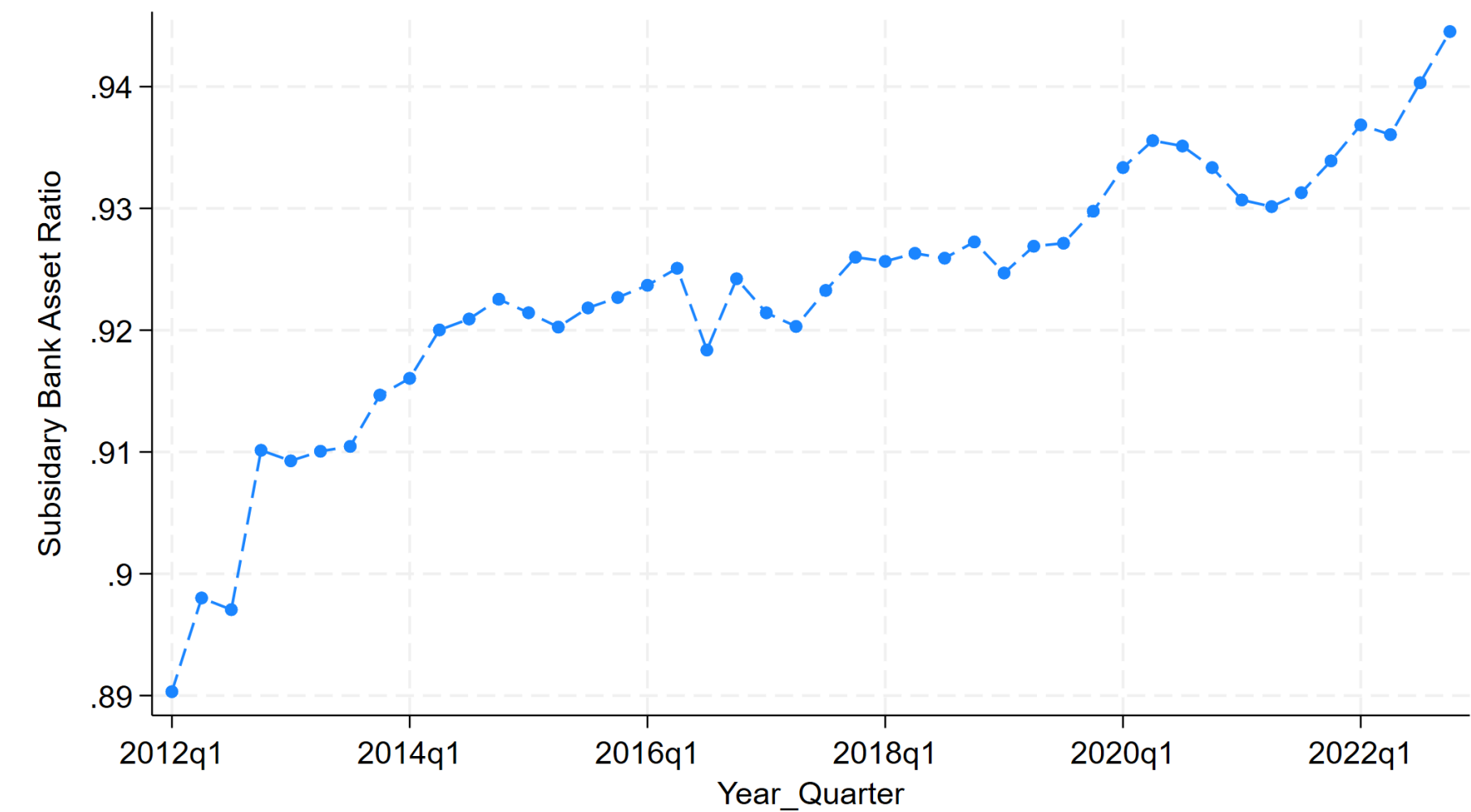
Data sources: Avraham et al. (2012)

Figure 2: Parent debt ratio over time



Source: FR Y-9C

Figure 3: Subsidiary bank’s asset ratio over time



Source: FR Y-9C and Call Reports

Appendix A: Proofs of Lemmas

Appendix A1: Proof of Lemma 1

Lemma 1. *If and only if $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \hat{\alpha}$, it holds that $\hat{D}'_p < \hat{D}''_p$. Excessive risk-taking occurs when $D_p < \hat{D}'_p$, whereas the first-best risk level emerges when $D_p \in [\hat{D}'_p, \hat{D}''_p]$.*

Proof. Based on Equations (9) and (10), when $S(1-\alpha) - D_b < 0$, the subsidiary bank fails, implying $D_p < \hat{D}'_p$. The BHC shareholder's payoff is:

$$L = P(R_b) [R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p] + (1 - P(R_b)) [R_n\alpha - D_n - D_p] \quad (1)$$

Applying Equations (6) and (7) and taking the first-order condition yields:

$$\begin{aligned} & -\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta(1-\alpha)(1-E_p - D_p) = 0 \\ \Rightarrow & 1 + \beta(1-E_p - D_p) = 2\beta R_b \\ \Rightarrow & R_b = \frac{1 + \beta(1-E_p - D_p)}{2\beta} \end{aligned}$$

When $S(1-\alpha) - D_b \geq 0$, the subsidiary bank remains solvent, implying $D_p \in [\hat{D}'_p, \hat{D}''_p]$. The BHC shareholder's payoff is:

$$L = P(R_b) [R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p] + (1 - P(R_b)) [R_n\alpha - D_n + S(1-\alpha) - D_b - D_p] \quad (2)$$

Taking the first-order condition yields:

$$\begin{aligned} & -\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta S(1-\alpha) = 0 \\ \Rightarrow & -\beta R_b + (1-\beta R_b) + \beta S = 0 \\ \Rightarrow & R_b = \frac{1 + \beta S}{2\beta} \end{aligned}$$

■

Appendix A2: Proof of Lemma 4

Lemma 2. *When both the parent/BHC and the subsidiary bank fail, excessive risk-taking still occurs because the limited liability distortion emerges at the subsidiary bank level.*

Proof. When the parent/BHC fails, market discipline is imposed on parent debt, and the interest rate on parent debt is r_p rather than the risk-free rate. The parent/BHC failure condition requires that the equity value of the nonbank subsidiary is insufficient to cover the parent's debt obligations:

$$R_n \alpha - \alpha(1 - E_p - D_p) - r_p D_p < 0 \quad (3)$$

The subsidiary bank failure condition requires that the subsidiary bank's assets do not cover its debt:

$$S(1 - \alpha) < (1 - \alpha)(1 - E_p - D_p) \quad (4)$$

The creditor's break-even condition is:

$$r_p D_p (1 - \beta R_b) + \beta R_b [R_n \alpha - \alpha(1 - E_p - D_p)] = D_p \quad (5)$$

The BHC shareholder's payoff is:

$$(1 - \beta R_b) [R_n \alpha + R_b(1 - \alpha) - (1 - E_p - D_p) - r_p D_p] + \beta R_b \cdot 0 \quad (6)$$

Substituting the creditor's break-even condition and rearranging yields:

$$(1 - \beta R_b) [R_n \alpha + R_b(1 - \alpha) - (1 - E_p - D_p)] - D_p + \beta R_b [R_n \alpha - \alpha(1 - E_p - D_p)]$$

Taking the first-order condition:

$$\begin{aligned} & -\beta R_b(1 - \alpha) + (1 - \beta R_b)(1 - \alpha) + \beta(1 - \alpha)(1 - E_p - D_p) = 0 \\ \Rightarrow & 1 + \beta(1 - E_p - D_p) = 2\beta R_b \\ \Rightarrow & R_b = \frac{1 + \beta(1 - E_p - D_p)}{2\beta} \end{aligned}$$

Since the default condition ensures $1 - E_p - D_p > S$, it follows that $R_b > R_b^{FB}$, confirming that excessive risk-taking occurs. ■

Appendix A3: Proof of Lemma 5

Lemma 3. *When the parent/BHC fails but the subsidiary bank remains solvent, the first-best solution obtains:*

$$R_b^{FB} = \frac{1 + \beta S}{2\beta} \quad (7)$$

Proof. The parent/BHC fails only when the subsidiary bank fails, which occurs with probability βR_b . Since market discipline is imposed on parent debt, the interest rate on parent debt is r_p .

The parent/BHC failure condition requires:

$$[R_n\alpha - \alpha(1 - E_p - D_p)] + [S(1 - \alpha) - (1 - \alpha)(1 - E_p - D_p)] - r_p D_p < 0 \quad (8)$$

The subsidiary bank solvency condition requires:

$$[R_n\alpha - \alpha(1 - E_p - D_p)] + [R_b(1 - \alpha) - (1 - \alpha)(1 - E_p - D_p)] - r_p D_p > 0 \quad (9)$$

The creditor's break-even condition is:

$$r_p D_p (1 - \beta R_b) + \beta R_b [R_n\alpha + S(1 - \alpha) - (1 - E_p - D_p)] = D_p \quad (10)$$

The BHC shareholder's payoff is:

$$(1 - \beta R_b) [R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p) - r_p D_p] + \beta R_b \cdot 0 \quad (11)$$

Substituting the creditor's break-even condition, the payoff simplifies to:

$$\begin{aligned} & (1 - \beta R_b) \left[R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p) - \frac{D_p - \beta R_b [R_n\alpha + S(1 - \alpha) - (1 - E_p - D_p)]}{1 - \beta R_b} \right] \\ &= (1 - \beta R_b) [R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p)] - D_p + \beta R_b [R_n\alpha + S(1 - \alpha) - (1 - E_p - D_p)] \\ &= R_n\alpha + (1 - \beta R_b) R_b(1 - \alpha) + \beta R_b S(1 - \alpha) - (1 - E_p - D_p) - D_p \end{aligned}$$

Taking the first-order condition:

$$\begin{aligned} & -\beta R_b(1 - \alpha) + (1 - \beta R_b)(1 - \alpha) + \beta S(1 - \alpha) = 0 \\ & \Rightarrow -\beta R_b + (1 - \beta R_b) + \beta S = 0 \\ & \Rightarrow R_b = \frac{1 + \beta S}{2\beta} \end{aligned}$$

When the parent/BHC fails but the subsidiary bank remains solvent, the absence of a limited liability distortion at the subsidiary bank level aligns the private risk-taking decision directly with the socially optimal first-best level. ■